

Autocrine BDNF-TrkB signalling within a single dendritic spine

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Brain-derived neurotrophic factor (BDNF) and its receptor TrkB are crucial for many forms of neuronal plasticity, including structural long-term potentiation (sLTP), which is a correlate of an animal's learning. However, it is unknown whether BDNF release and TrkB activation occur during sLTP, and if so, when and where. Here, using a fluorescence resonance energy transfer-based sensor for TrkB and two-photon fluorescence lifetime imaging microscopy, we monitor TrkB activity in single dendritic spines of CA1 pyramidal neurons in cultured murine hippocampal slices. In response to sLTP induction, we find fast (onset < 1 min) and sustained (>20 min) activation of TrkB in the stimulated spine that depends on NMDAR (N-methyl-d-aspartate receptor) and CaMKII signalling and on postsynaptically synthesized BDNF. We confirm the presence of postsynaptic BDNF using electron microscopy to localize endogenous BDNF to dendrites and spines of hippocampal CA1 pyramidal neurons. Consistent with these findings, we also show rapid, glutamate-uncaging-evoked, time-locked BDNF release from single dendritic spines using BDNF fused to superecliptic pHluorin. We demonstrate that this postsynaptic BDNF-TrkB signalling pathway is necessary for both structural and functional LTP. Together, these findings reveal a spine-autonomous, autocrine signalling mechanism involving NMDAR-CaMKII-dependent BDNF release from stimulated dendritic spines and subsequent TrkB activation on these same spines that is crucial for structural and functional plasticity.